

A Two-Stage Earthquake Event Classification Model Based on Diffusion Probability Model

Fanchun Meng^{ID}, Tao Ren^{ID}, Pengyu Wang^{ID}, Xinliang Liu^{ID}, Wenjuan Xiang, and Xinyu He

Abstract—Rapid and accurate classification of earthquake (eq) events is a serious challenge in seismology and disaster mitigation. Problems, such as data imbalance, model interpretability, and model generalization, limit the application of artificial intelligence methods in this research area. This article introduces a real-time two-stage diffusion eq event classification (DiffEEC) model based on the diffusion probability model (DPM). DiffEEC uses a two-stage classification approach and combines DPM and knowledge distillation (KD) techniques. DiffEEC focuses on seismic phase-related features and source mechanism-related features by combining time-series features extracted by convolutional layers, DPM output, and InSAR data features to better extract core seismic data information and reduce reliance on manual feature design. DiffEEC uses focal loss to solve the data imbalance problem. Thus, DiffEEC can address data scarcity and imbalance, feature acquisition and selection, variability and complexity of seismic event processing, and model generalization through the mechanism. Experiments show that DiffEEC performs better in eq event classification (EC).

Index Terms—Diffusion probability model (DPM), earthquake (eq) event classification (EC), feature fusion, knowledge distillation (KD).

I. INTRODUCTION

EARTHQUAKE (eq) event classification (EC) is a challenging problem in eq monitoring [1], [2], [3]. Distinguishing tectonic and nontectonic eqs is often distinguished by manual and semiautomatic techniques. Manual techniques require extensive seismic data review, resulting in significant workload and human fault-tolerance limitations. Semiautomatic approaches use feature extraction and machine learning to distinguish tectonic and nontectonic seismic sources [3], [4].

Moreover, the machine learning model in this task has become more sophisticated [5], [6]. Lyubushin et al. [7]

designed a linear Bayesian discriminator using multifractal singularity spectra that measured the fractal dimension of time moments. Li et al. [8] used a generative adversarial network for eq prediction. Lindenbaum et al. [9] recommended neural networks and deep canonical correlations for autonomous discrimination. A signal-to-noise ratio assessor and a short-time/long-time mean detector were used to identify eq triggers. Bergman et al. [10] developed an array-based seismic discrimination methodology using diffusion maps, preprocessing time–frequency wave analysis.

Deep learning methodologies offer several benefits among various techniques. Convolutional neural network (CNN) accurately distinguishes minor eqs from artificial noise, showing robust deep learning capabilities [11], [12], [13], [14]. Supervised clustering is an efficient feature representation for targeted clustering [15]. Feature fusion strategies increase interpretability in seismic event discrimination [16]. Recurrent neural networks can efficiently process seismology data by integrating with CNNs, extracting sequence characteristics [17]. Kong et al. [18] used deep learning and physical features to detect explosions eps and eqs.

However, challenges to the classification of eq events are listed as follows.

- 1) Seismic data are affected by instrumentation, the environment, and signal attenuation, limiting reliable interpretation of the classification model.
- 2) Nontectonic data are much less than tectonic, posing significant challenges to the accuracy and generalizability of the model.
- 3) Seismic data require transformation into features in other dimensions, but current methods may miss essential seismic data information.
- 4) Classifying methods may be inaccurate due to difficulty in handling different focal mechanisms, crustal structures, and seismic processes.

We propose a real-time two-stage diffusion eq EC (DiffEEC) model based on the diffusion probability model (DPM), a probabilistic model that may refer to techniques for modeling data distribution and can be used in various applications, innovated as follows.

- 1) The noise addition process of the DPM can be considered data augmentation, allowing the model to learn differences in signal characteristics, thereby reducing noise interference.
- 2) DiffEEC learns features using knowledge distillation (KD), which is a technique for transferring knowledge

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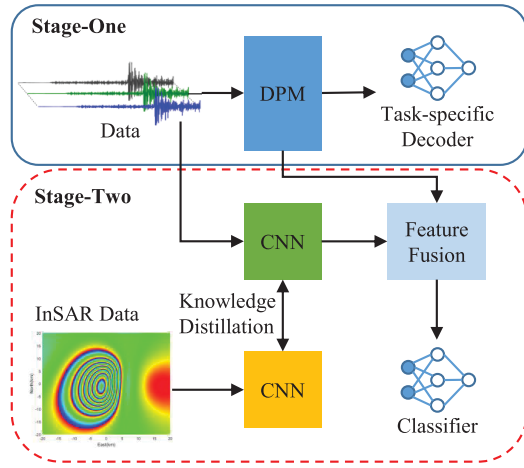


Fig. 1. Flowchart of the proposed method and network structure.

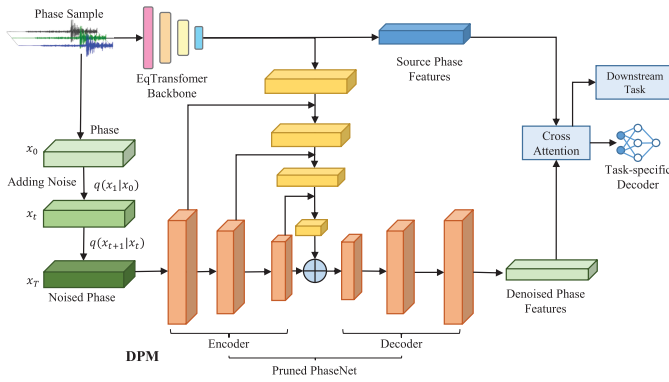


Fig. 2. Overall architecture of DPM.

from a complex model to a more efficient one to improve recognition and reduce reliance on human design.

- 3) DiffEEC constructs a local framework and interconnections of seismic data for accurate handling of seismic characteristics, enhancing adaptability to diverse seismic situations.
- 4) DiffEEC uses focal loss [19] to handle unbalanced seismic datasets, improving classification performance for specific categories and increasing accuracy.

This article contains five sections, including an introduction section. Section II describes the details of our proposed method. The experiments are performed and presented with relevant discussions in Section III. Section IV gives the conclusion.

II. METHODOLOGY AND DATA

A. Framework Overview

Referred to in Fig. 1, DiffEEC includes two stages: phase features extraction (PFE) stage (detailed in Fig. 2) and EC stage (detailed in Fig. 4). The DPM optimizes seismic phase features through task-dependent decoding and postprocessing to verify their validity. EC uses PFE model inference results and uses coded seismic phase features.

InSAR data can obtain seismic source mechanism information for classification of seismic events. KD is used to transfer

complex InSAR knowledge to CNN for performance improvement. The CNN extracted time-series features are fused with DPM output to focus on seismic phase features, which are also important for seismic EC. The classifier finalizes EC from the feature fusion output.

B. Phase Features Extraction

1) *Diffusion Probability Model*: Fig. 2 details the implementation process of the DPM algorithm. Seismic phase features are extracted using EqTransformer [20], and PhaseNet [21] is used for feature fusion and inference. The model uses only PhaseNet parameters for real-time inference in seismic phase pickup. Pruning technology [22] optimizes PhaseNet, reducing nonessential parameters and leading to faster computational speed and accuracy. In the scene setting, the noise x_T is randomly generated according to the standard normal distribution $\mathcal{N}(0, I)$. According to the principle of diffusion modeling, it will gradually transform the noise x_T into the original sample x_0 . The variance of the variable at stage t ($0 \leq t \leq T$) is denoted by β_t .

The forward process of the conditional diffusion model is

$$q(x_t|x_{t-1}) \sim \mathcal{N}(x_t; \sqrt{1 - \beta_t}x_{t-1}, \beta_t I). \quad (1)$$

For convenience, we denote a series of constant

$$\alpha_t = 1 - \beta_t, \bar{\alpha}_t = \prod_{s=1}^t \alpha_s. \quad (2)$$

The noisy sample at step t is transformed from the input data x_0 by

$$q(x_t|x_{t-1}) \sim \mathcal{N}(x_t; \sqrt{\bar{\alpha}_t}x_{t-1}, (1 - \alpha_t)I) \quad (3)$$

$$x_t \sim \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1 - \bar{\alpha}_t}\epsilon, \epsilon \sim \mathcal{N}(0, I). \quad (4)$$

$\sum_{\theta}(x_t, t)$ is the covariance predictor, and $\epsilon_0(x_t, t)$ is the denoising model. In our experiments, PhaseNet is used as the denoising network. In the denoising process, the diffusion model progressively refines the noisy sample x_t . The reverse diffusion process is written as follows:

$$p_{\theta}(x_t|x_{t-1}) \sim \mathcal{N}(x_{t-1}; \mu_{\theta}(x_t, t), \sum_{\theta}(x_t, t)). \quad (5)$$

The process is similar to the data enhancement method of adding noise to samples.

2) *Cross-Attention Module*: After obtaining the output of the conditional diffusion model, we design a cross-attention module (CA) to refine the original phase feature, as shown in Fig. 3. Specifically, the output of the conditional diffusion model is treated as the source of K and V , while the original phase feature from the perspective view transformer is projected into Q . The cross-attention process of the two-stream features is formulated as follows:

$$CA(Q, K, V) = \text{Attn}(QW_i^Q, KW_i^K, VW_i^V)W^{\text{Out}} \quad (6)$$

$$\text{Attn}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V. \quad (7)$$

Q , K , and V are linearly mapped to calculate the attention matrix Attn . Then, the refined phase feature is obtained from the output layer W^{Out} , which aims to facilitate the downstream tasks to learn better.

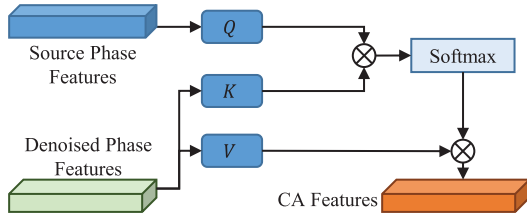


Fig. 3. Structure of the CA.

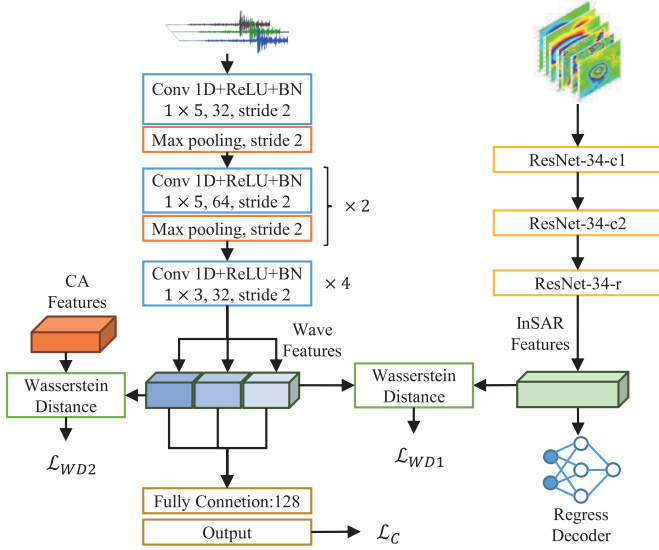


Fig. 4. Flowchart of KD process in Stage-Two.

3) *Training Loss: Diffusion Loss*: We denote the Gaussian noise at time step t as $\bar{z}_t$. The diffusion loss $\mathcal{L}_{\text{diff}}$ is defined as follows:

$$\mathcal{L}_{\text{diff}} = \mathbb{E} \left[\left\| \bar{z}_t - \sum_{\theta} \left(\sqrt{\alpha_t} x_0 + \sqrt{1 - \alpha_t} \bar{z}_t, t \right) \right\|^2 \right]. \quad (8)$$

Task-Specific Training Loss: Here, we still use cross-entropy loss function $\mathcal{L}_{\text{phase}}$ of PhaseNet. By balancing the two loss functions [23], the overall loss function $\mathcal{L}_{\text{One}}$ of Stage-One is

$$\mathcal{L}_{\text{One}} = \frac{1}{2\sigma^2} \mathcal{L}_{\text{phase}} + \mathcal{L}_{\text{diff}} + \log \sigma \quad (9)$$

where σ is the variance of the PhaseNet prediction distribution.

C. Event Classification

1) *Phase Picking*: Using PhaseNet as the denoising net in DPM, it is used to detect the seismic phase with the aid of the rapid earthquake association and location (REAL) [24] method for correlation. Due to varying epicentral distances, the duration of the usable waveform varies by station. This range is typically tens of seconds to several minutes. Therefore, we use a 100-s waveform post the initial break to ensure its overall integrity.

2) *Knowledge Distillation*: The stage integrates KD and classification, as illustrated in Fig. 4. For time-series data, referring to previous research [13], we use CNN for signal preprocessing, proportionately distributing extracted features, merging them with CA features and InSAR data features

derived from InSAR data [25] via KD. After blending, these features reconnect to the original features and proceed to the fully connected layer for final classification. This KD section facilitates CNN's learning of seismic phase and source mechanism features, thus enhancing classification accuracy.

InSAR data features significantly contribute to the seismic EC process, amplifying the model's predictive capacity. This technology uses small ground surface deformations (including vertical and horizontal displacements, and deformation gradients) to retrieve seismic event data. These deformation data are invaluable for identifying seismic event characteristics and magnitude. The model extracts these deformation features from InSAR data for EC. Augmenting InSAR data with seismic waveform information aids the model's understanding of seismic event characteristics and, thus, enhances classification accuracy. InSAR data features can be combined with time-series features gleaned through the convolutional layer and DPM output. This feature fusion bolsters core seismic event information extraction, mitigates dependence on manual feature design, and escalates the model's generalization aptitude in seismic EC. Seismic source mechanism is a critical asset for seismic EC, and InSAR data provide insights into the seismic source mechanism, such as seismic fracture surface geometry, sliding direction, and stress release pattern. This information paints a detailed picture of seismic events and aids the model in classifying them with improved accuracy. Using InSAR data features, the model can holistically understand seismic events and obtain rich classification information. This advance in predictive capability and accuracy will significantly enhance seismic hazard monitoring and early warning.

3) *Training Loss: Classification Loss*: Due to the unbalanced distribution of different types of event samples (e.g., more blasting events in mining areas and fewer other events), focal loss is used to overcome the two major challenges of imbalanced category samples and difficult sample mining. Focal loss is a modification of the standard cross-entropy loss function that assigns higher weights to misclassified examples from the minority class, thereby focusing the model's attention on those difficult examples. By doing so, focal loss effectively reduces the impact of the dominant class and improves the model's ability to generalize across different classes. The loss function $\mathcal{L}_C$ used for classification is

$$\mathcal{L}_C = (1 - p_t)^\gamma \log(p_t) \quad (10)$$

where p_t is the category probability value of the outcome predicted by the model and $\gamma = 0.25$ is the focusing parameter, which increases the value of γ and increases the network's sensitivity to misclassified observations.

Normalized Wasserstein Distance (NWD) Loss: Due to the differences in the feature distributions obtained from various feature extraction methods and the imbalance of sample distribution that may lead to the emergence of outliers, we choose to use the Wasserstein distance as the loss function for KD. It is worth emphasizing that because of its continuity and smoothness, the Wasserstein distance makes it easier to find a solid solution during the optimization process, effectively reducing the instability caused by too much change in the gradient. CA, wave, and InSAR features are all third-order

feature tensors that can be converted to a 2-D Gaussian distribution.

Let $v_1 \sim \mathcal{N}_1(m_1, C_1)$ and $v_2 \sim \mathcal{N}_2(m_2, C_2)$ be two normal distributions on $\mathbb{R}^n$, with respective expected values m_1 and $m_2 \in \mathbb{R}^n$ and symmetric positive semidefinite covariance matrices C_1 and $C_2 \in \mathbb{R}^{n \times n}$. Then, with respect to the usual Euclidean norm on $\mathbb{R}^n$, the 2-Wasserstein distance between v_1 and v_2 is

$$W_2^2(v_1, v_2) = \|m_1 - m_2\|_2^2 + \text{Tr}(C_1 + C_2 - 2(C_1^{1/2} C_1 C_2^{1/2})^{1/2}). \quad (11)$$

It can be simplified as follows:

$$W_2^2(v_1, v_2) = \|m_1 - m_2\|_2^2 + \|C_1^{1/2} - C_2^{1/2}\|_F^2 \quad (12)$$

where $C^{1/2}$ denotes the principal square root of C and $\|\cdot\|_F$ is Frobenius norm. Thus, NWD is

$$\text{NWD}(\mathcal{N}_1, \mathcal{N}_2) = \exp\left(-\frac{\sqrt{W_2^2(v_1, v_2)}}{2\Sigma^2}\right) \quad (13)$$

where Σ is the expected value of the sample data for the batch. Obviously, NWD is the similarity measure, and the corresponding loss function $\mathcal{L}_{\text{WD}}$ is

$$\mathcal{L}_{\text{WD}} = 1 - \text{NWD}(\mathcal{N}_1, \mathcal{N}_2). \quad (14)$$

The overall loss function $\mathcal{L}_{\text{Two}}$ of Stage-Two is

$$\mathcal{L}_{\text{Two}} = \mathcal{L}_C + \mathcal{L}_{\text{WD1}} + \mathcal{L}_{\text{WD2}}. \quad (15)$$

In this phase, the network that extracts the CA features with the InSAR features performs only the inference process. The parameters of the backbone CNN [13] are adjusted during backpropagation.

III. RESULTS AND DISCUSSION

A. Data

The data come from the Comprehensive Dataset of Chinese Seismic Network (CSNCD) [26], a database of the China Earthquake Networks Center (CENC), covering 1.3 million events between January 2009 and December 2022. The dataset includes seven event categories: volcanotectonic eqs (ves), suspected events (ses), others (ots), subsidence (ss), suspected explosions (sps), eps, and eqs; 99.7% are eqs, while the rest fall into the remaining six categories, as detailed in Table I and Fig. 5. InSAR data are sourced from the National Earthquake Data Center, with simulated data generated using CSNCD labels using the method outlined in [25] to supplement the dataset. Data are divided into training (80%), validation (10%), and testing (10%) partitions.

B. Results

We evaluate DiffEEC on the test set by precision (Pr), recall (Re), F1-score (F_1), and accuracy (ACC). Among them, the evaluation metrics are calculated as $\text{Pr} = \text{TP}/(\text{TP} + \text{FP})$, $\text{Re} = \text{TP}/(\text{TP} + \text{FN})$, $F_1 = (2 \times \text{Pr} \times \text{Re})/(\text{Pr} + \text{Re})$, and $\text{ACC} = (\text{TP} + \text{TN})/(\text{TP} + \text{FP} + \text{TN} + \text{FN})$, where TP, TN, FP, and FN are the total number of true positive, true negative,

TABLE I
NUMBERS OF DIFFERENT EVENT TYPES

Type	Percentage(%)	Quantity
volcano-tectonic earthquake	0.3	2
suspected event	4.8	20
other type	10.3	41
subsidence	21.2	84
suspected explosion	26.1	103
explosion	32.8	129

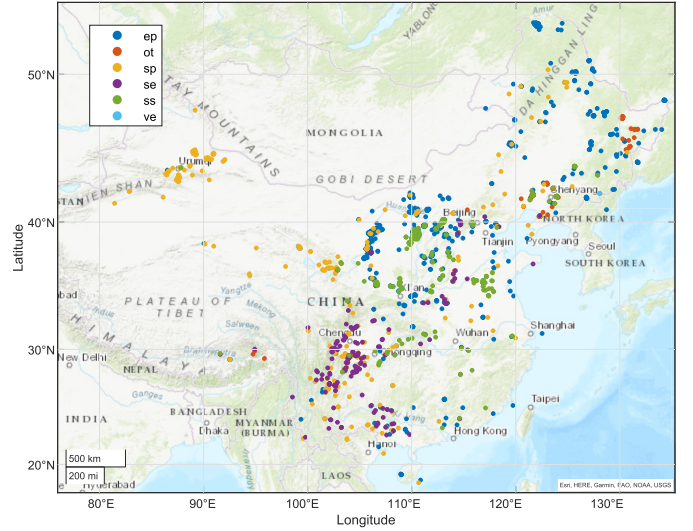


Fig. 5. Distribution of event epicenter locations.

TABLE II
DIFFEEC TESTS RESULTS ON DIFFERENT TYPES OF EVENTS

Type	Re	Pr	F_1
earthquake	0.9600	1.0000	0.9796
volcano-tectonic earthquake	1.0000	0.0054	0.0108
suspected event	1.0000	0.0006	0.0012
other type	0.7500	0.0166	0.0324
subsidence	0.8750	0.0063	0.0125
suspected explosion	0.5000	0.0198	0.0382
explosion	0.4615	0.0242	0.0460

false positive, and false negative, respectively, considering ground truth as positive.

Since several stations record a seismic event, simultaneous analysis of different records from different stations is used to isolate seismic events, ensuring greater accuracy. Adhering to the principle of majority voting, the records of all stations classifying the incident dictate the type of incident. In the case of a draw, the event type is determined through three rounds of DiffEEC interaction. Events classified with most records are considered event types. Fig. 6 illustrates the confusion matrix of experimental results using the test set. Table II shows the performance of the model.

Fig. 6 and Table II show that the model delivers 96% accuracy on the test set. Furthermore, by implementing focal loss and KD configurations within the network, the model rarely incorrectly labels the event as eq. In addition, the model's erroneous predictions mainly pertain to misclassifying eq for the remaining six events. This problem is caused by an overabundance of samples in the eq category coupled with

eq	124426	183	175	1100	242	238	3246
ve		1					
ot			3	1			
ss			1	7			
sp			1		5	4	
ep	1		1		5	6	
se							2
	eq	ve	ot	ss	sp	ep	se

Fig. 6. Confusion matrix for classification results.

frequent spatial and temporal proximities to other categories. For example, samples misclassified as *ve* in the *eq* category, located near the Changbai Mountains, and mainly microseisms detected by the Jilin network, are examined. The number of misclassified *se* samples, classified as *sp* or *ep*, is relatively high, as experts use different criteria while manually labeling. In addition, the majority of these samples are microseismic. The model's inadequate classification of *sp* and *ep* could also be attributed to mainly small eqs with similar source mechanisms, hindering effective differentiation in their characteristics.

C. Comparison Experiment

We reproduce the results of the comparison methods on the same dataset and compare them with DiffEEC, as shown in Table III. On the CSNCD dataset, DiffEEC's F_1 score stands at 0.9793, demonstrating improvement over other methods (up to 0.9517). This indicates that DiffEEC classifies and predicts more precisely with the CSNCD dataset. For the ENSN dataset, DiffEEC's F_1 score exceeds that of CapsuleNN, yet its ACC is lower, indicating superior overall classification accuracy for DiffEEC on the ENSN dataset, enhancing its domain generalization skills. On the SEISDMC dataset, DiffEEC's F_1 score is 0.9811, the highest of all methods, while its ACC score of 0.9677 is also the highest. This represents DiffEEC's performance on the SEISDMC dataset, providing the highest classification accuracy. In addition, DiffEEC, trained on CSNCD data nationwide in China, demonstrated its adaptability to the SEISDMC dataset, reflecting its ability to identify features unique to different geologies. Conjunction with Fig. 6, we observe that most DiffEEC errors are concentrated in the *ss* and *se* classes, indicating its strong generalization performance in the plains region.

D. Ablation Experiment

To determine whether the relevant structures and functions within the model are critical to system functionality, we

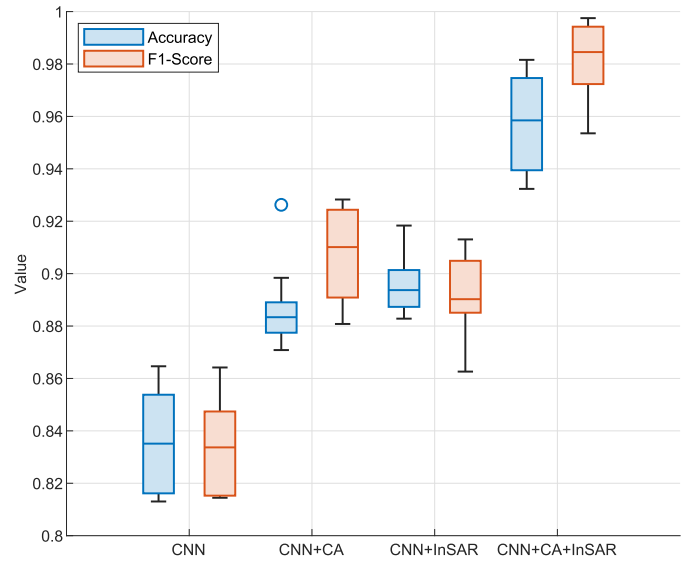


Fig. 7. Boxplot of the distribution for F1-score and accuracy results for the ablation experiment.

eliminate the KD processes of CA and InSAR features in EC, respectively. The related experimental results are illustrated in Table IV. To ensure consistency, the model training and testing process is performed ten times, as shown in Fig. 7. Notably, the incorporation of CA features gives a considerable boost to F1-score, while InSAR features have less impact. This indicates that CA features play a vital role in classification, providing quality information for categorizing diverse objects. DiffEEC successfully combines the two features, showing significant improvements. This highlights the potential for synergies between multiple feature sources to enhance the performance of the classification model. Combining CNN, CA, and InSAR features allows us to extract information from multiple angles and describe samples in a comprehensive manner, boosting accuracy.

E. Robustness Experiments

Due to errors and technical challenges in DiffEEC's antecedent functions, such as seismic phase detection, correlation, and localization, the impact of station data on EC was examined. Station result voting determines the types of events; therefore, assessments covering scenarios with different epicenter distances for the same event were conducted. As can be seen in Fig. 8, DiffEEC's accuracy steadily improves with more participating stations and epicenter distances. Notably, the increase in accuracy between 125 and 150 km is well in line with China's initial motion polarity trend, validating DiffEEC's findings.

Second, station results were randomly invalidated for the same event to measure their effect on event accuracy. The inefficiency index was determined as the proportion of failed stations over the total number of recording stations. For example, if ten stations record an event but three fail, the inefficiency is 30%. These failed stations are excluded from model inference. Fig. 9 shows that event accuracy decreases as inefficiency progresses. The reason for this is that the

TABLE III
EXPERIMENTAL RESULTS OF DIFFEEC AND COMPARATIVE METHODS

Model	F_1	ACC	Speed (Obs/sec) ¹	Test set
SVM+Polynomial [6]	0.8604	0.8353	917	CSNCD
SVM+RBF [6]	0.9459	0.9207	963	CSNCD
NB [6]	0.8494	0.8098	812	CSNCD
LR [6]	0.6284	0.6633	2257	CSNCD
RLR [6]	0.9157	0.8847	2038	CSNCD
Bagging [4]	0.7454	0.7218	1774	CSNCD
SVM [13]	0.6003	0.5634	899	CSNCD
CNN [13]	0.8427	0.8405	634	CSNCD
LeNet [14] ²	0.9517	0.9432	877	CSNCD
CapsuleNN	0.9163	0.9608	415	ENSN
SVM2	0.9253	0.9235	963	SEISDMC
XGboost	0.9650	0.9512	1875	SEISDMC
DiffEEC	0.9793	0.9599	634	CSNCD
DiffEEC	0.9357	0.9532	634	ENSN
DiffEEC	0.9811	0.9677	634	SEISDMC

¹ The hardware environment is: CPU Intel Core i7-13700K, 64G memory, two Nvidia RTX 4080 SUPER 16 GB GDDR6X GPUs. Obs/sec represents the number of observed samples per second.

² Use only earthquake and volcano-tectonic earthquake data.

TABLE IV
RESULTS OF ABLATION EXPERIMENTS

Model	F_1	ACC
CNN	0.8427	0.8405
CNN+CA Features	0.9002	0.8984
CNN+InSAR Features	0.8902	0.8981
CNN+CA Features+InSAR Features	0.9793	0.9599

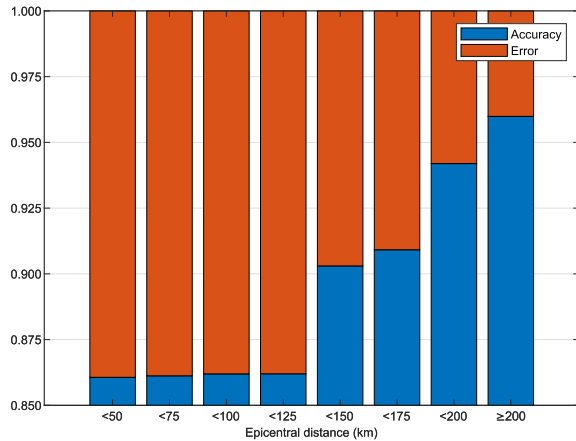


Fig. 8. Effect of stations with different epicenter distances on EC results.

station network for microseismic events is sparse, resulting in significant inefficiencies even with one or two station failures, leading to potential misclassified eq events. Despite this, DiffEEC maintains over 0.9 accuracy at inefficiencies below 30%, showing superior robustness.

F. Feature Sensitivity Analysis

Gradient weighted class activation mapping (Grad-CAM) methodology [27] is used to identify key predictive features in deep learning decisions. It focuses on extracting feature maps from the final convolutional layer, and in our tests, we use post-KD features. By calculating the gradient of final category scores with respect to these feature maps, we can discern the individual pixels for the final classification. A typical

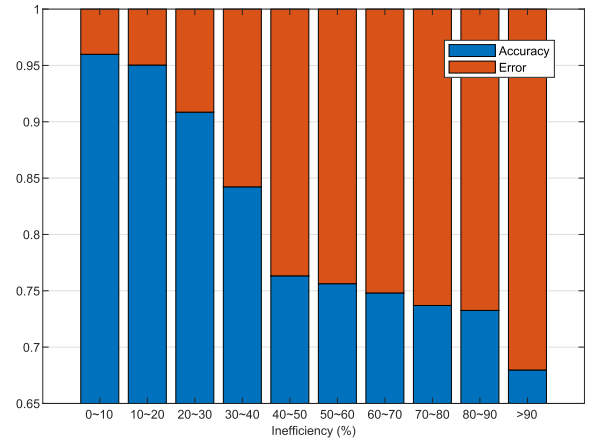


Fig. 9. Impact of station inefficiency on network results.

Grad-CAM heatmap, as illustrated in Fig. 10, transparently demonstrates how distinct time-series zones affect classification results. According to seismological understanding and experience, the P-S phase often reveals particular differences due to the unique energy release pattern of nontectonic activities, such as eps, ss, or landslides, as compared with tectonic eqs. Nontectonic and tectonic seismic waveforms exhibit distinct surface wave characteristics. Fig. 10 indicates the model's pronounced sensitivity to the seismic phase and the postsurface wave phase, which aligns with seismological theories and experiences, proving further the efficiency of KD operation for the seismic phase and source mechanism.

G. Optimization Analysis

To achieve a faster and more accurate computation speed in a real-time system, we evaluate DiffEEC postpruning and quantization.

1) *Pruning*: In practical use, due to the KD mechanism, PhaseNet is used to capture seismic phases in PFE, and our backbone CNN completes the classification in EC. For direct end-side application without data upload, pruning and

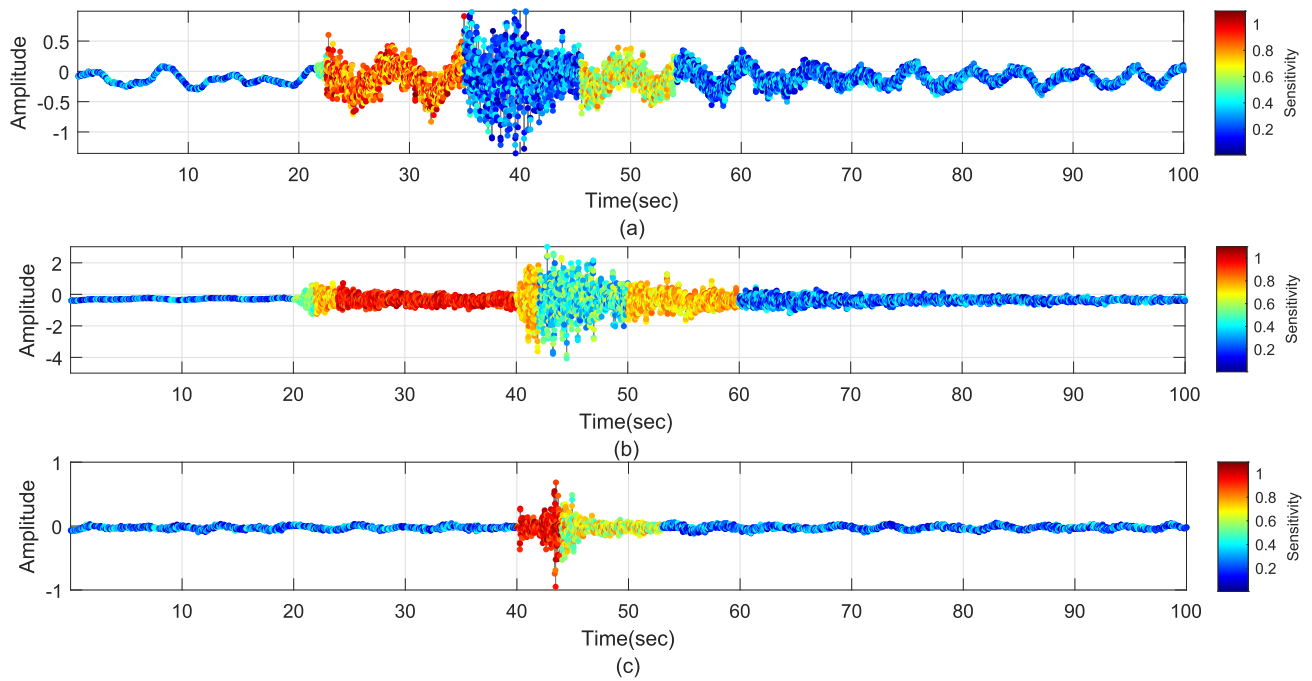


Fig. 10. Grad-CAM on a time series recorded. (a) ep recorded on December 9, 2017 at 22:44:41 on AKS station, Xinjiang. (b) se recorded on March 12, 2016 at 18:15:30 on WNQ station, Xinjiang. (c) eq recorded on January 31, 2019 at 13:28:29 on MXI station, Sichuan.

quantization were applied to streamline the backbone CNN, maintaining accuracy while increasing efficiency. We pruned the network according to [22], achieving an accuracy of 0.9582 at a pruning rate of 50%.

2) *Quantization*: For the pruned network, 8-bit quantization, such as quantization-aware training [28] and post-training quantization [29], converts model weights from 32-bit FP32 to 8-bit INT8, reducing model size by 75% while maintaining 0.9477 accuracy.

We validate DiffEEC's integration into ESPRH [30] eq warning system, maintaining precision levels consistent with quantification. This demonstrates that our methodology enhances eq warning system performance without compromising accuracy. Integrating DiffEEC models can lead to additional computational complexity, requiring a blend of real-time performance and accuracy, and decisions based on specific application contexts and requirements.

H. Discussion

DiffEEC is successful in classifying seismic events, but there are still limitations that require discussion and refinement.

- 1) The DiffEEC model uses focal loss to solve the problem of data imbalance, but it does not fully correct it. While focal loss allows the model to focus on certain categories, in severe circumstances, samples from these few categories may lack sufficient data for full classification. Therefore, the efficiency of the DiffEEC model could be hindered in highly skewed seismic datasets.
- 2) Despite KD's use to enhance feature learning and reduce reliance on manual design, details can be lost in the KD process. KD improves performance by transferring knowledge from complex models to simplified ones,

but some critical information could be compromised, impacting model performance or limiting its ability to adapt to certain scenarios.

- 3) The DiffEEC model may falter in addressing various source mechanisms, crustal structures, seismic processes, and seismic EC. Classification is affected by many factors, such as diversity and complexity of source mechanisms. Although the DiffEEC model refocused on seismic phase characteristics through methods, such as feature fusion and sequence feature extraction, it can still face classification errors or an inability to accurately discern different seismic events for complex events.

Future research can augment seismic EC by leveraging additional data sources and sophisticated deep learning techniques.

- 1) Incorporate seismic data into seismic EC with other associated data sources, such as geophysical data, surface deformation data, meteorological data, and so on. Leveraging information from multiple sources can enhance the accuracy and dependability of seismic EC.
- 2) Consider diverse types and characteristics of seismic data, and develop adaptive deep learning models to manage these disparate data, for example, amalgamating multimodal data, such as time-series data, image data, and text data for seismic EC.
- 3) Experiment with unsupervised learning methods for seismic EC to uncover hidden patterns and structures in the data. This can help us classify and understand seismic events at a more refined level while reducing reliance on labeled data.
- 4) Employ migration learning and domain adaptation techniques to adapt the existing seismic EC models to new seismic datasets or seismic data from diverse regions. Through migration learning, training time and sample

requirements on new datasets can be reduced, and the model's adaptability can be enhanced. These are potential avenues for future seismic EC research that can enhance the accuracy, resilience, and usability of seismic EC using complementary data sources and advanced deep learning methods.

IV. CONCLUSION

This article proposes a two-stage seismic EC model called DiffEEC using DPM, which can obtain high-quality classification results. This two-stage model can preprocess seismic data to improve the stability of the classification model. In feature extraction, DiffEEC effectively extracts vital information and reduces dependence on human-designed features. In addition, by adopting focal loss to solve the problem of data imbalance, the classification performance of a few categories is optimized, and the overall classification accuracy is improved. To enhance the generalization capability of the model, KD techniques are employed to transfer knowledge of complex InSAR data to the time-series-based model, which improves performance. Finally, with the synergy of feature fusion and classifiers, DiffEEC can accurately classify seismic events with an accuracy of 0.9599. In summary, DiffEEC performs better in classifying seismic events. Future research can continue to expand and improve on this foundation to meet emerging challenges and needs in seismic EC.

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